

Roll No.

--	--	--	--	--	--	--

193

50

End Semester Examination June 2024

Name of the Course: B.Tech ECE

Semester: VIII

Name of the Paper: Testing for VLSI Circuits

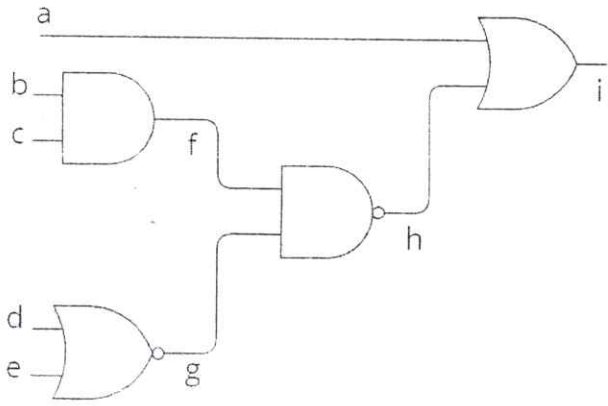
Paper Code: **TPE 014**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO1
(a)	Explain the role of testing in VLSI circuits and testing philosophy.	
(b)	Explain the following terms: a. Fault coverage b. Equivalence fault c. Yield d. Rule of Ten	
(c)	Differentiate between verification and testing in VLSI circuits and describe the system level operation.	
Q2	(20 marks)	CO2
(a)	Describe delay test and its classification in detail. What is the need for delay test?	
(b)	Describe controllability and observability in detail. Write down the expressions for flip flop controllability and observability.	
(c)	Explain Boolean difference method of generating test vectors in detail. Generate the test vector considering stuck-at-0 fault at node "f". 	
Q3	(20 marks)	CO3
(a)	Explain the following terms: a) D-intersection. b) Primitive D-cube of fault (PDCF) c) Propagation D-cube (PDC) d) Singular cover	
(b)	Explain coupling fault and its types in detail.	
(c)	Explain transition fault along with its state diagram.	
Q4	(20 marks)	CO4,
(a)	Explain delay test and its types in detail.	

91

(b)	Describe the block diagram of memory architecture and explain stuck-at-faults in memory testing.	CO6
(c)	Explain scan architecture and technique, also discuss its disadvantages.	
Q5	(20 marks)	CO5
(a)	Describe BIST architecture and its benefits.	
(b)	Differentiate between structural versus functional test.	
(c)	Explain the difficulties in testing sequential circuit.	

A

Roll No.

TPE-015

B. TECH. (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, June, 2024

WIRELESS SENSOR NETWORKS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the basic principle of wireless Ad hoc networks. (CO1)
(b) Explain “Hybrid” and “Mesh Networks” of ad hoc wireless networks. Point at least five differences between cellular networks and ad hoc wireless networks. (CO1)
(c) What are the challenges of WSN ? Explain all in brief. (CO1)
2. (a) Explain the “Hidden Terminal” and “Exposed Terminal” problems in MANET. (CO2 & CO3)
(b) Explain the following application of WSNs : (CO2 & CO3)
 - (i) Industrial Applications

P. T. O.

- (ii) Environmental Applications
 - (iii) Seismic Activity monitoring
 - (iv) Greenhouse Monitoring Applications
 - (c) Explain the features, advantages and disadvantages of Zigbee Protocols.
(CO2 & CO3)
3. (a) What is the “wake-up scheme” in MAC ? How NAV (Network Vector Allocation) works to avoid collision.
(CO3 & CO4)
- (b) Explain the B- MAC (Berkeley’s MAC) protocol in detail.
(CO3 & CO4)
- (c) Explain the following :
(CO3 & CO4)
- (i) MACA protocol
 - (ii) Chain-Based Three Level Scheme
4. (a) What is the “Data Dissemination” process in WSN ? Differentiate between “Flooding” and “Gossiping” data dissemination process.
(CO4)
- (b) Explain the differences between the following architectures in WSN :
(i) Layered Architecture (ii) Clustered Architecture. (CO4)
- (c) Explain the “Binary Scheme” with example for data gathering process.
(CO4)
5. (a) What are the hardware. components of WSN5 ? Explain each one in detail.
(CO5 & CO6)
- (b) What are the Gateway concept in WSN ? Why gateways are required in WSNs ?
(CO5 & CO6)
- (c) How and why WSN5 to internet communication process is advantageous ?
(CO5 & CO6)

Roll No.

TPE-008

B. TECH. (ECE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, June, 2024
NEURAL NETWORK AND MACHINE LEARNING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are Biological Neural Networks ? Describe some attractive features of the biological neural network that make it superior to the most sophisticated Artificial Intelligence computer system for pattern recognition tasks. (CO1, CO2)

(b) Explain Linear Regression method with a suitable example. (CO1, CO2)

(c) Explain Logistic Regression method with a suitable example.

(CO1, CO2)

2. (a) Explain McCulloch-Pitts model of Neurons.

(CO2)

P. T. O.

(2)

- (b) Design a pattern classifier using basic neural network model with S as a set of patterns to be classified into classes P and Q : (CO2)

$S = \{0000, 0001, 0010, 0011, 0100, 0101, 0110, 0111\}$

$P = \{0000, 0001, 0010, 0100\}$

$Q = \{0011, 0101, 0110, 0111\}$

- (c) Explain the structure of Perceptron. Implement AND Operation using the Perceptron. (CO2)
3. (a) Illustrate the steps involved in a pattern classification task. (CO3)
- (b) Illustrate how support vector machines are used for pattern classification. (CO3)
- (c) Explain any two activation functions. (CO3)
4. (a) Explain the Kohonen's self-organizing map and its significance. (CO4)
- (b) Explain Bidirectional Associative Neural Network. (CO4)
- (c) Explain Hetro-Associative Neural Network. (CO4)
5. (a) Explain Pattern Storage and Retrieval. (CO5, CO6)
- (b) Explain the back propagation training algorithm in Neural Network. (CO5, CO6)
- (c) Explain the architecture of recurrent neural network. (CO5, CO6)

TPE-008

191

End-Semester Examination 2024

Name of the course: B. Tech (ECE)

Semester: VIII

Name of the Paper: CMOS Analog Circuit Design

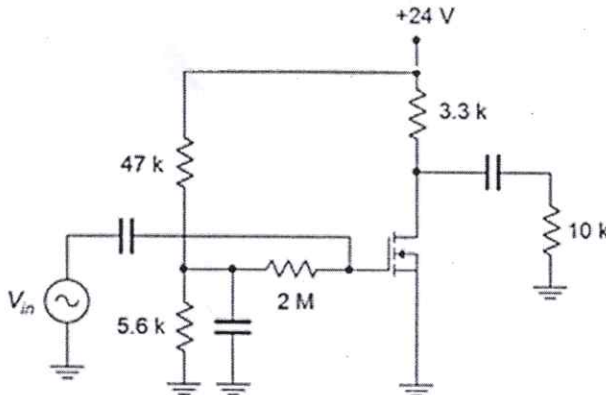
Paper Code: **TPE 016**

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two **questions** among a, b, & c **in each main question**.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(20 marks)	CO1
(a)	Explain current voltage characteristics of second order effects in MOS device.	
(b)	Explain DC and AC analysis of small signal model MOS transistor.	
(c)	Explain the all the modes of operation of MOSFET.	
Q2	(20 marks)	CO2
(a)	Derive expression for gain in current source load CS MOSFET amplifier.	
(b)	For the circuit of Figure, determine the voltage gain and input impedance. Assume $V_T = 2$ V, $I_{D_S} = 50$ mA at $V_{GS} = 5$ V.	
		
(c)	For the circuit of Figure, determine the voltage gain and input impedance. Assume $V_{GS} = -0.75$ V and $I_{DSS} = 6$ mA.	

Q3	(20 marks)	CO3
(a)	Explain the difference between single ended and differential operation.	
(b)	Explain the concept of common mode response of differential amplifier.	
(c)	Explain in brief about low power amplifier.	
Q4	(20 marks)	CO4
(a)	What do you understand about cascade current mirror?	
(b)	Describe about frequency response of single stage amplifier.	
(c)	Explain in brief difference about one stage op-amp and two stage op-amp.	
Q5	(20 marks)	CO5/6
(a)	Write all the properties of feedback circuits.	
(b)	How MOS can be used as a unity gain buffer?	
(c)	Explain Industrial scope of CMOS analog circuit.	